ADAPTIVE VEHICLE DASHBOARD SIMULATOR

-Based on LPC2129 Microcontroller

GROUP-10

V25BE10K7
V25BE10P8

CONTENTS

- Introduction
- Project Objectives
- Software Requirements
- Hardware Requirements
- Peripheral Configurations
- Data Field: Bit Packing
- Block Diagram
- Node - A Execution Flow
- External Interrupt
- Node - B Execution Flow
- Results and Demonstration
- Future Scope
- Conclusion

INTRODUCTION

The Adaptive Vehicle Dashboard Simulator is an embedded system designed to demonstrate vehicle parameter monitoring and communication between electronic control units. It uses two LPC2129 microcontrollers connected through the CAN protocol to demonstrate reliable data exchange, sensor monitoring, and vehicle indicator control.

PROJECT OBJECTIVES

- Establish reliable 125 kbps CAN communication between two LPC2129 microcontrollers with interrupt-driven reception.
- Successfully acquire temperature, speed, and distance data using sensors and detect headlight and indicator inputs.
- Implement efficient bit-level packing and unpacking of vehicle data within a 64-bit CAN payload.
- Display real-time vehicle telemetry through UART at 9600 bps and control the corresponding headlight and indicator LEDs.

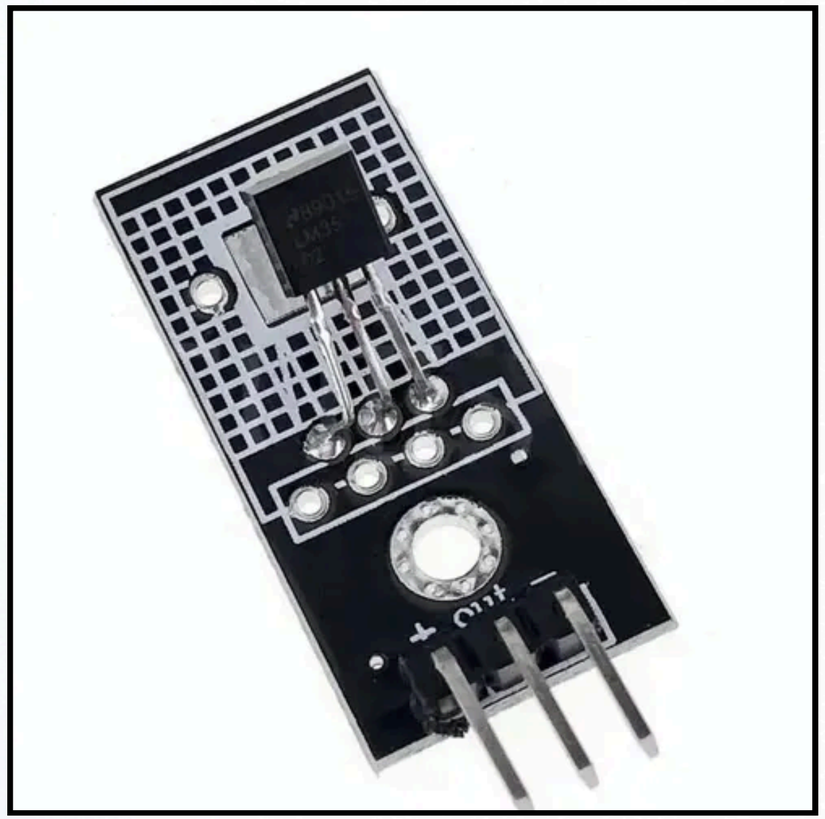
SOFTWARE REQUIREMENTS

- Keil µVision 5
- Flash Magic
- Serial Terminal
- Embedded C

HARDWARE REQUIREMENTS

Component	Quantity	Purpose
LPC2129 ARM7 Development Board	2	Transmitter and receiver
LM35 Temperature Sensor	1	Temperature measurement
Potentiometer	1	Simulate vehicle speed input
HC-SR04 Ultrasonic Sensor	1	Distance measurement
CAN Transceiver Module	2	CAN signal conversion

LM35 TEMPERATURE SENSOR



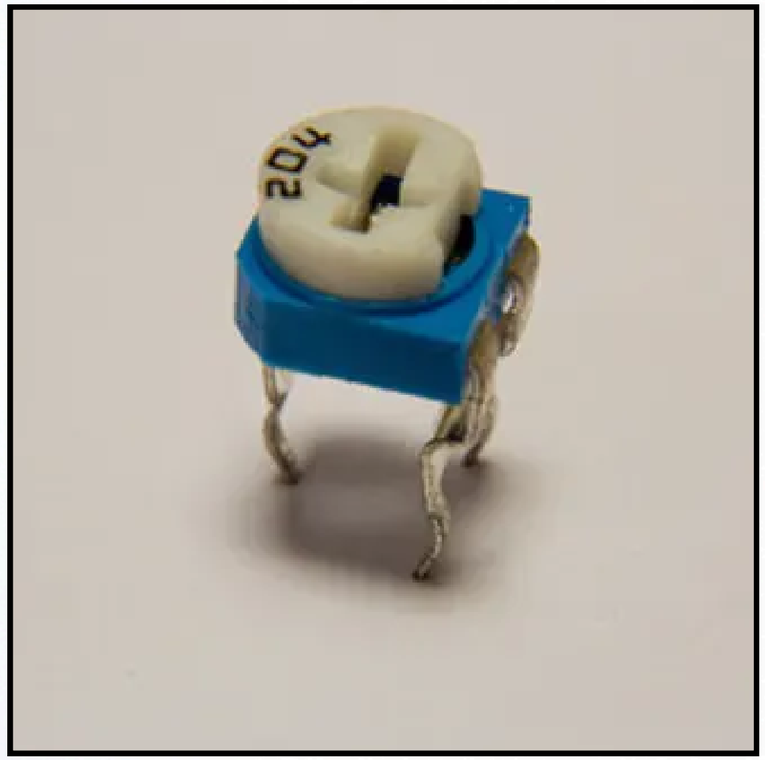
Specifications:

Output Range : –55°C to +150°C

Operating Voltage: 4V–30V

Accuracy : ±0.5°C (at 25°C)

POTENTIOMETER



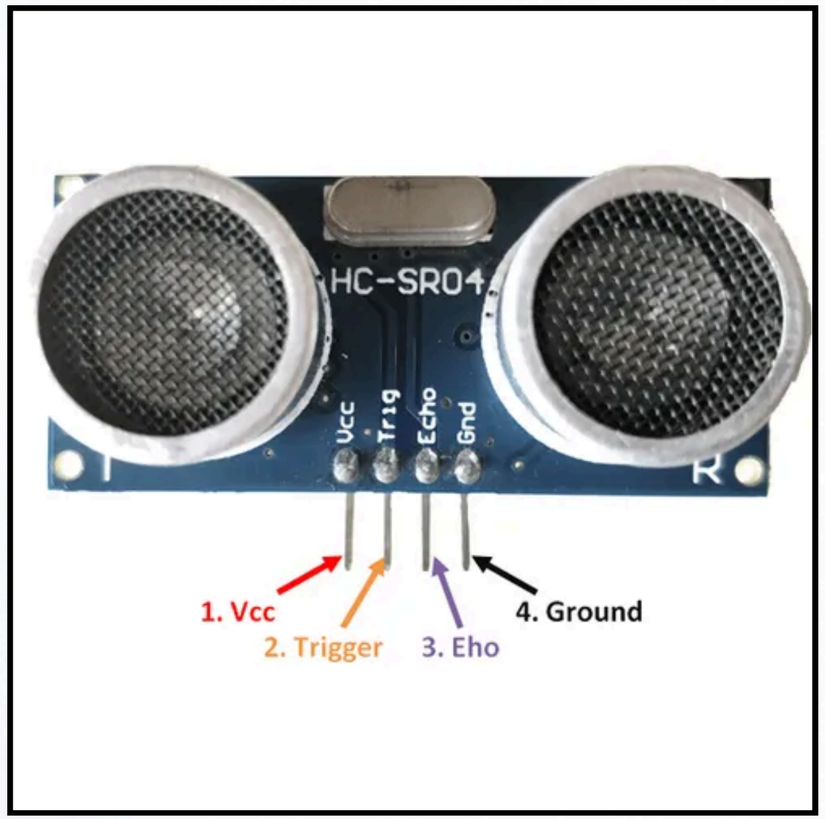
Specifications:

Resistance : 10kΩ

Output :analog voltage(0–3.3V)

Resolution : 10-bit ADC

HC-SR04 ULTRASONIC SENSOR



Specifications:

Measurement Range: 2–400cm

Operating Voltage : 3.3V DC

Resolution : 0.3cm

UART Communication Configuration

Configuration	Value
UART Peripheral	UART0
Peripheral Clock	60 MHz
Baud Rate	9600 bps
Data Bits	8 bits
Parity	None
Stop Bits	1 bit

ADC Configuration

Configuration	Value
ADC Peripheral	ADC0
Channels Used	2 (AD0.0, AD0.1)
Resolution	10 bits
Digital Levels	1024 (0–1023)
Reference Voltage	3.3V
Output Format	10-bit Digital Value

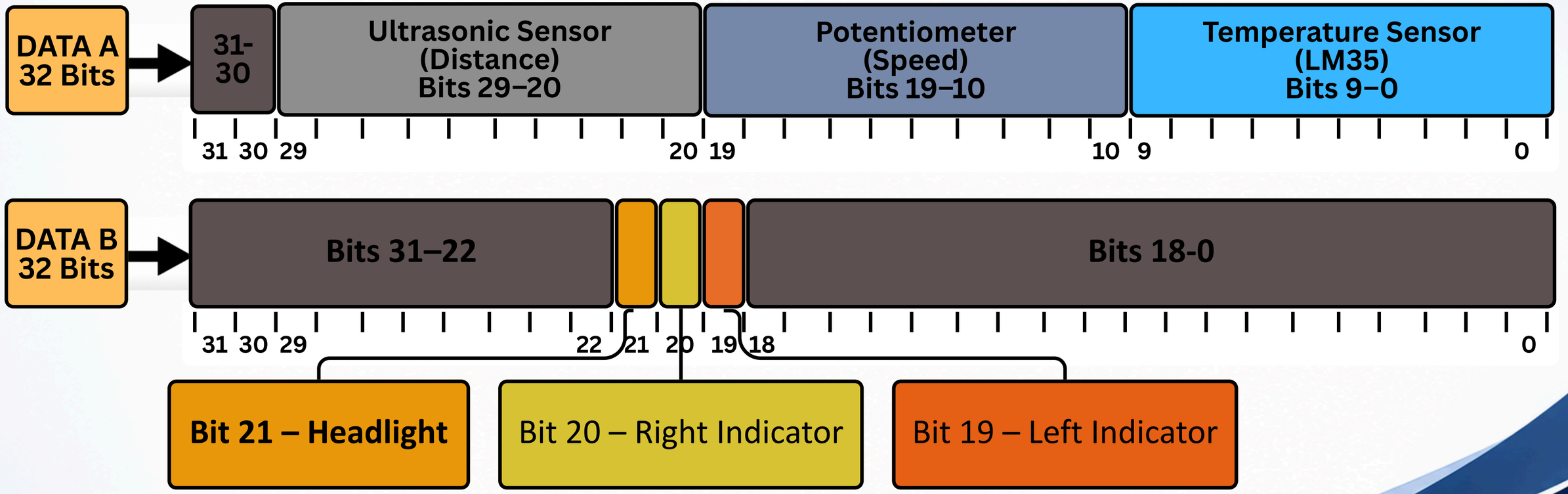
External Interrupt Configuration

Configuration	Value
Interrupt Type	External Interrupt
Interrupt Sources	EINT0, EINT1, EINT2
Trigger Mode	Edge Triggered
Trigger Edge	Falling Edge
Interrupt Controller	VIC

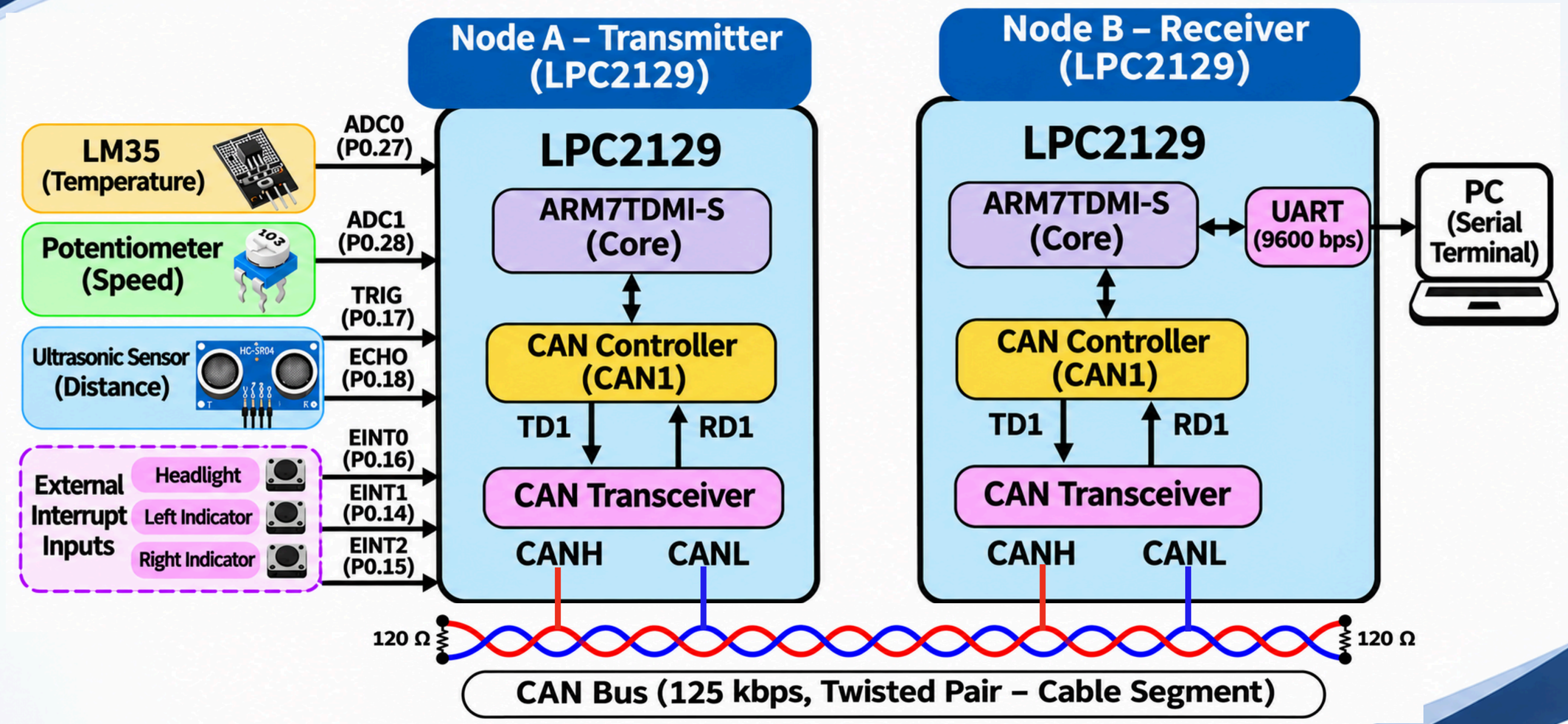
CAN1 Communication Configuration

Configuration	Value
Baud Rate	125 kbps
Peripheral Clock	60 MHz
Acceptance Filter Mode	0x02
Frame Format	Standard (11-bit ID)
Payload	8 bytes

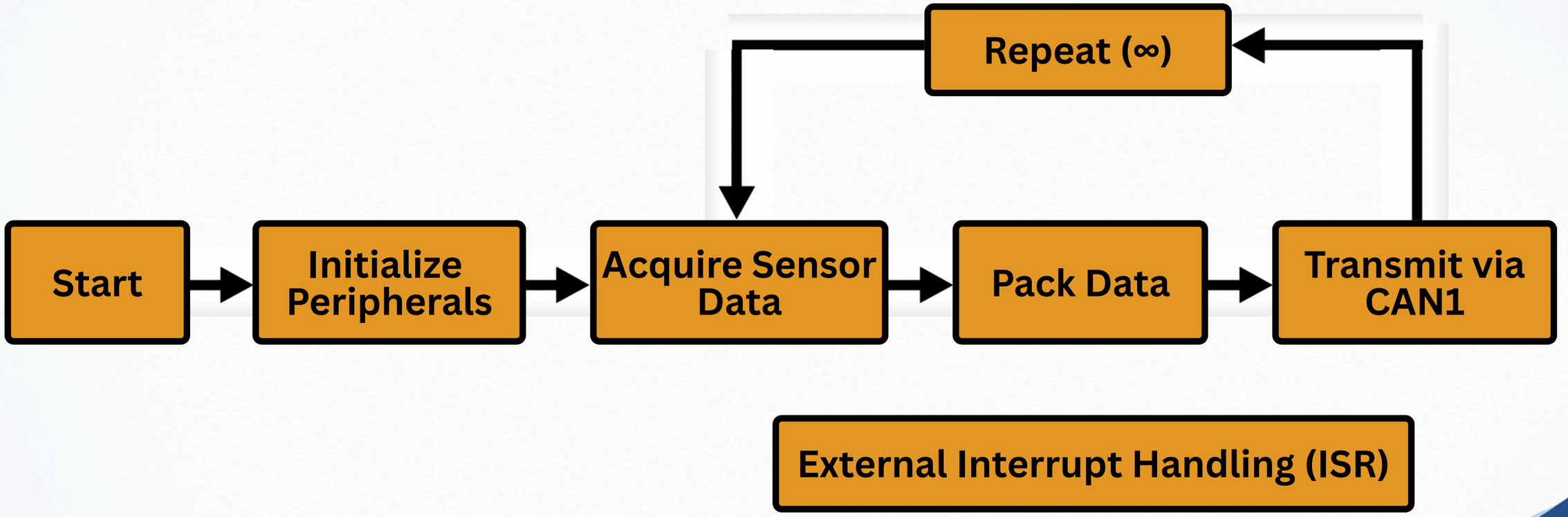
Data Field: Bit Packing



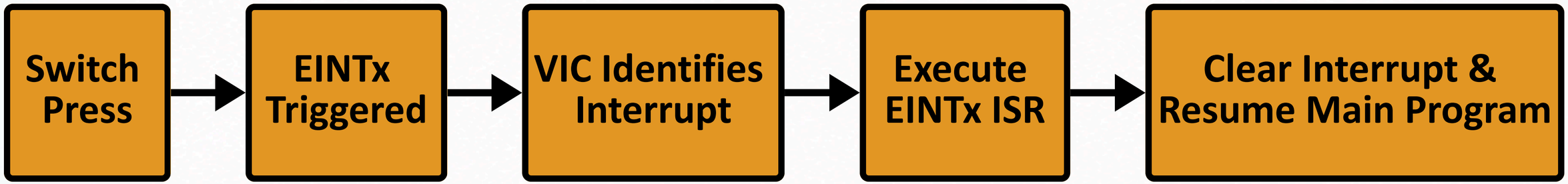
BLOCK DIAGRAM



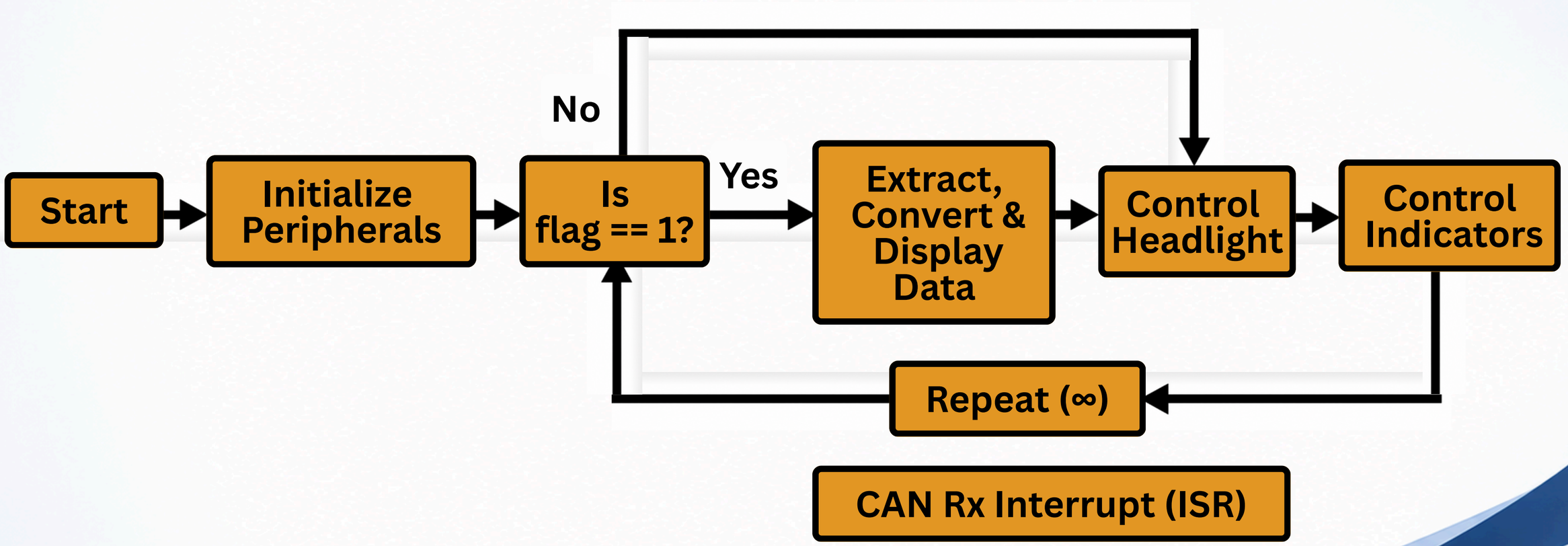
NODE - A
EXECUTION FLOW



EXTERNAL INTERRUPT

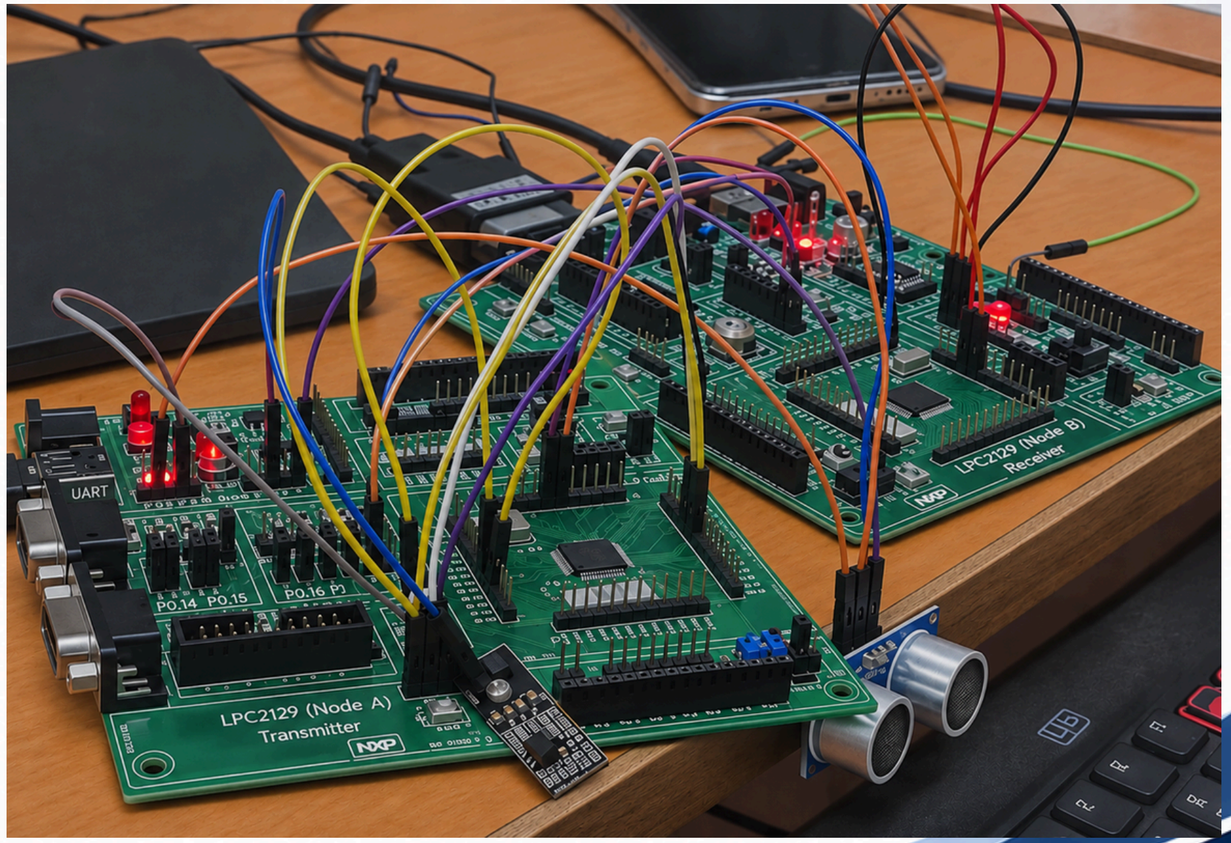


NODE - B
EXECUTION FLOW



RESULTS AND DEMONSTRATION

```
left indic ON
head light ON
TEMPERATURE: 27.21474536
SPEED: 0.0km/h
FUEL LEVEL: 0.0L
head light ON
<-  .
head light ON
TEMPERATURE: 28.64516C
SPEED: 0.0km/h
FUEL LEVEL: 0.0L
    .
head light ON
TEMPERATURE: 30.1C
SPEED: 0.0km/h
FUEL LEVEL: 0.0L
    .
head light ON
TEMPERATURE: 28.709676C
SPEED: 0.0km/h
FUEL LEVEL: 0.0L
    . ->
right indic ON
head light ON
TEMPERATURE: 31.290323C
SPEED: 0.0km/h
FUEL LEVEL: 0.0L
    . ->
```



FUTURE SCOPE

- **Expanded CAN Network:** Connect additional control units to monitor and exchange data across multiple vehicle subsystems.
- **Intelligent Warning System:** Introduce automatic alerts for abnormal temperature, low fuel and unsafe vehicle conditions.
- **Enhanced Fault Detection:** Implement diagnostic mechanisms to detect sensor failures, CAN communication errors and abnormal vehicle parameter readings.

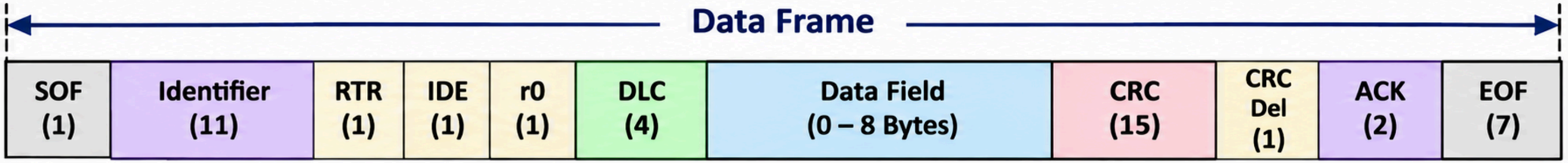
CONCLUSION

- Successfully implemented an adaptive vehicle dashboard simulator using the LPC2129 microcontroller.
- Interfaced sensors and input switches to monitor vehicle parameters and driver inputs.
- Established CAN communication between two LPC2129 nodes for reliable data exchange.
- Implemented interrupt-based input handling and indicator control.
- Displayed the received information through UART for real-time monitoring.

Thank You

CAN PROTOCOL
DATA FRAME STRUCTURE

Standard Data Frame (11-bit Identifier)



Field	Description
SOF	Start of Frame – Dominant bit (0) to indicate frame start.
Identifier	11-bit message identifier (CAN 2.0A, Standard).
RTR	Remote Transmission Request bit (0 = Data frame, 1 = Remote frame).
IDE	Identifier Extension bit (0 = Standard 11-bit, 1 = Extended 29-bit).
r0	Reserved bit (always 0).

Field	Description
DLC	Data Length Code – Number of data bytes (0 to 8).
Data Field	Application data (0 to 8 bytes).
CRC	15-bit Cyclic Redundancy Check for error detection.
CRC Del	CRC Delimiter – Recessive bit (1).
ACK	Acknowledge field – ACK slot and ACK delimiter (2 bits).
EOF	End of Frame – 7 consecutive recessive bits (1).